AI-Powered Customer Support Bot with Automated Drift Detection

A production-grade observability project built on
Respan · RAG · LLM-as-Judge Evals · MLOps

Tech Stack: Respan · Groq LLM · ChromaDB · LangChain · HuggingFace

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Respan Organization ID: 7d80e5ff-a37a-49ce-97d1-7e0cbc17c7a7

Executive Summary

This project demonstrates a production-grade AI customer support system that combines Retrieval-Augmented Generation (RAG) with automated quality evaluation and real-time drift detection — addressing one of the most critical challenges in deploying LLMs at scale: ensuring sustained response quality over time.

Problem Statement

Production AI systems face silent quality degradation over time. Help documentation becomes outdated, model behavior shifts, and retrieval quality drifts — all without any visible error. Teams only discover failures when users complain. This project solves that by implementing a self-monitoring AI pipeline that catches degradation automatically.

Solution

The system integrates three layers: a RAG pipeline for grounded, accurate answers; an LLM-as-judge evaluation framework scoring every response across three quality dimensions; and a drift detection engine that monitors rolling quality trends and fires alerts when thresholds are breached. All LLM calls are fully traced via Respan.

Key Results

Metric	Value	Description
Evaluation Coverage	**100%**	Every response scored automatically
Eval Dimensions	**3 judges**	Faithfulness, Relevance, Resolution
Avg Quality Score	**0.86 / 1.0**	Across 7 demo evaluations
Drift Threshold	**0.70**	Alert fires below this rolling average
LLM Calls Traced	**57**	Full observability in Respan dashboard
Total Cost	**$0.00**	Groq free tier + local HuggingFace embeddings
Respan Org ID	**7d80e5ff...c7a7**	**7d80e5ff-a37a-49ce-97d1-7e0cbc17c7a7**

System Architecture

The system is composed of four sequential layers. Each layer is independently runnable, testable, and observable via Respan.

01 Document Ingestion

Help documents (.txt / .md) are loaded, split into 500-character overlapping chunks, and embedded using HuggingFace all-MiniLM-L6-v2 (runs locally, no API cost). Vectors persist in ChromaDB.

- Input: Raw .txt / .md help documents
- Embedding: HuggingFace all-MiniLM-L6-v2 (local, free)
- Storage: ChromaDB vector store
- Output: 42 searchable chunks ready for retrieval

02 RAG Pipeline + Respan Tracing

User queries are embedded and matched against ChromaDB to retrieve the top-4 most relevant chunks. An augmented prompt is built and sent to Groq via Respan Gateway. Every call is traced with latency, cost, and token counts.

- Retrieval: Top-4 chunks via cosine similarity
- LLM: groq/llama-3.1-8b-instant via Respan Gateway
- Fallback: Direct Groq API if gateway fails
- Tracing: Full span tree logged to Respan dashboard

03 LLM-as-Judge Evaluation

Three independent LLM judge prompts score each response on a 0.0-1.0 scale using Groq (free tier). Scores are saved to data/scores.json for longitudinal drift analysis.

- Faithfulness: Is the answer grounded in retrieved context?
- Relevance: Did retrieved chunks match the query?
- Resolution: Was the user problem actually solved?
- All scores persisted with timestamps for trend analysis

04 Drift Detection & Monitoring

A rolling 7-day average is computed on every run. If it drops below 0.70, a drift alert fires with remediation steps. Exit code 1 on drift makes it CI/CD-compatible.

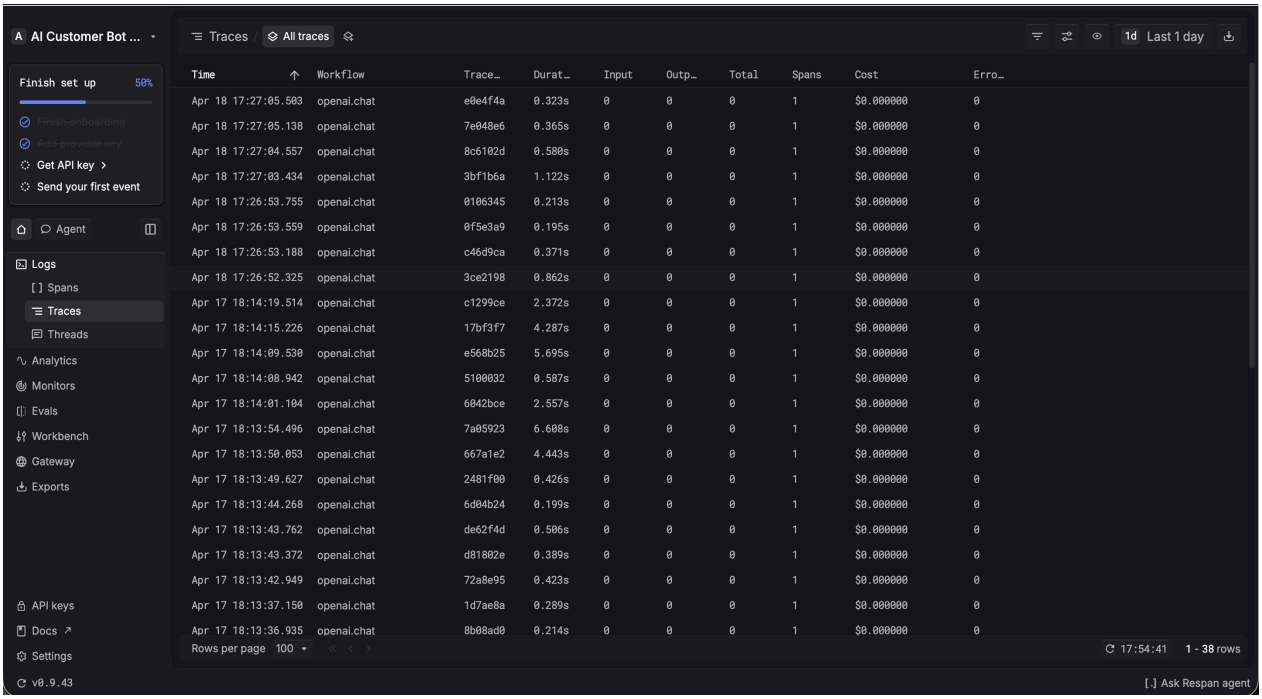
- Rolling window: 7-day configurable average
- Alert threshold: 0.70 (configurable in monitor.py)
- Trend analysis: Compares last 3 vs previous 3 evaluations
- Exit code 1 on drift — compatible with CI/CD pipelines

Respan Observability Dashboard

All LLM calls made by the support bot are fully traced and visible in the Respan dashboard. The following screenshots show live observability data captured during the project demo run.

Trace View — Full Request History

Every query generates a trace showing workflow name, duration, token counts, spans, and cost. The dashboard captured 38 traces across the demo session.



Time	Workflow	Trace…	Durat…	Input	Outp…	Total	Spans	Cost	Erro…
Apr 18 17:27:05.503	openai.chat	e0e4f4a	0.323s	0	0	0	1	$0.000000	0
Apr 18 17:27:05.138	openai.chat	7e048e6	0.365s	0	0	0	1	$0.000000	0
Apr 18 17:27:04.557	openai.chat	8c6102d	0.580s	0	0	0	1	$0.000000	0
Apr 18 17:27:03.434	openai.chat	3bf1b6a	1.122s	0	0	0	1	$0.000000	0
Apr 18 17:26:53.755	openai.chat	0106345	0.213s	0	0	0	1	$0.000000	0
Apr 18 17:26:53.559	openai.chat	0f5e3a9	0.195s	0	0	0	1	$0.000000	0
Apr 18 17:26:53.188	openai.chat	c46d9ca	0.371s	0	0	0	1	$0.000000	0
Apr 18 17:26:52.325	openai.chat	3ce2198	0.862s	0	0	0	1	$0.000000	0
Apr 17 18:14:19.514	openai.chat	c1299ce	2.372s	0	0	0	1	$0.000000	0
Apr 17 18:14:15.226	openai.chat	17bf3f7	4.287s	0	0	0	1	$0.000000	0
Apr 17 18:14:09.530	openai.chat	e568b25	5.695s	0	0	0	1	$0.000000	0
Apr 17 18:14:08.942	openai.chat	5100032	0.587s	0	0	0	1	$0.000000	0
Apr 17 18:14:01.104	openai.chat	6042bce	2.557s	0	0	0	1	$0.000000	0
Apr 17 18:13:54.496	openai.chat	7a05923	6.608s	0	0	0	1	$0.000000	0
Apr 17 18:13:50.053	openai.chat	667a1e2	4.443s	0	0	0	1	$0.000000	0
Apr 17 18:13:49.627	openai.chat	2481f00	0.426s	0	0	0	1	$0.000000	0
Apr 17 18:13:44.268	openai.chat	6d04b24	0.199s	0	0	0	1	$0.000000	0
Apr 17 18:13:43.762	openai.chat	de62f4d	0.506s	0	0	0	1	$0.000000	0
Apr 17 18:13:43.372	openai.chat	d81802e	0.389s	0	0	0	1	$0.000000	0
Apr 17 18:13:42.949	openai.chat	72a8e95	0.423s	0	0	0	1	$0.000000	0
Apr 17 18:13:37.150	openai.chat	1d7ae8a	0.289s	0	0	0	1	$0.000000	0
Apr 17 18:13:36.935	openai.chat	8b08ad0	0.214s	0	0	0	1	$0.000000	0

Figure 1: Respan Traces view — 38 logged LLM calls with latency, tokens, and cost per request

Spans View — Individual Call Details

The Spans view drills into each individual LLM call with log IDs, customer identifiers, thread IDs, and span names — enabling precise debugging of any individual request in production.

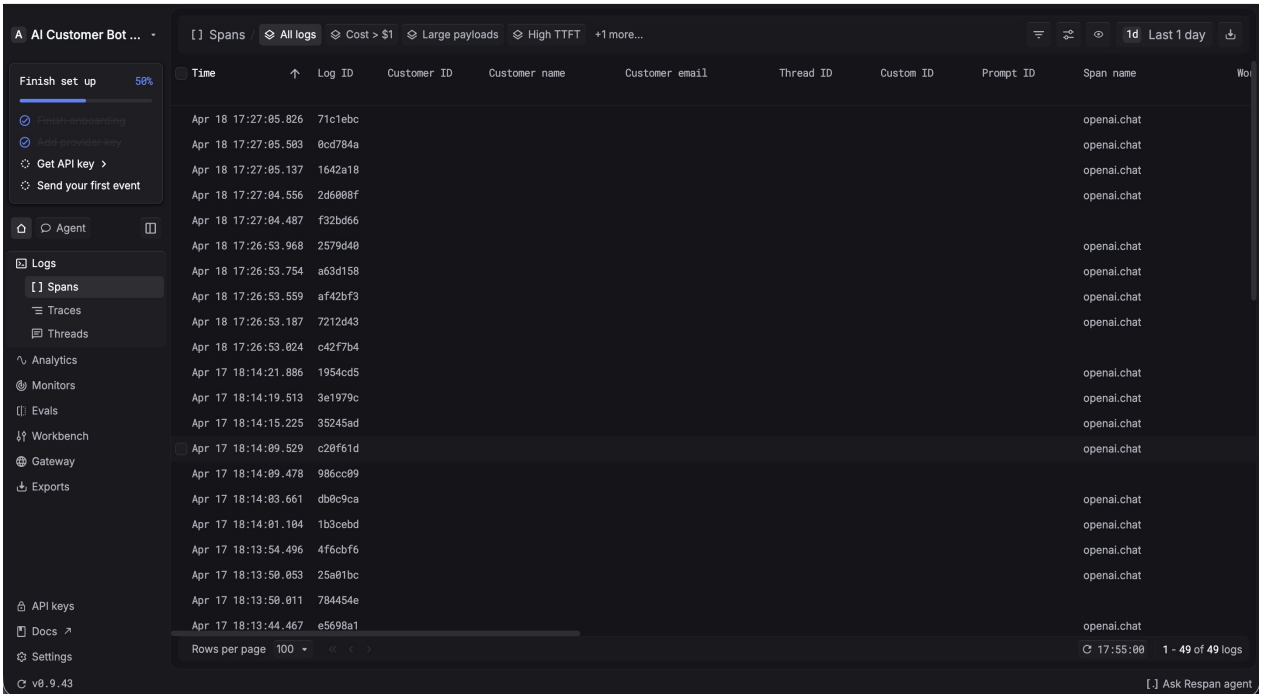


Figure 2: Respan Spans view — 49 individual call logs with full metadata

Analytics and Metrics Dashboard

The analytics dashboard provides a comprehensive view of system health, model performance, and cost trends across the entire demo session.

Figure 3: Respan Analytics — 57 total requests, model breakdown, latency percentiles, and token usage

Metric	Value	Notes
Total Requests	**57**	Across demo and evaluation runs
Error Rate	**7.02%**	Caused by initial OpenAI quota errors (resolved)
Total Cost	**$0.00**	Groq free tier — zero spend
Input Tokens	**7,985**	All prompt and context tokens
Output Tokens	**650**	All generated response tokens
Generation p50	**0.36s**	Median response time
Generation p95	**5.70s**	95th percentile latency
Primary Model	**groq/llama-3.1-8b-instant**	Via Respan Gateway

Evaluation Results and Drift Detection

LLM-as-Judge Score Summary

The following table shows evaluation scores across all 7 demo runs. Rows 6 and 7 demonstrate the system detecting quality degradation when retrieval returns insufficiently relevant chunks.

#	Question	Faith.	Relev.	Resol.	Avg	Status
1	How do I reset my password?	1.00	1.00	1.00	1.00	Pass
2	How do I cancel my subscription?	1.00	1.00	1.00	1.00	Pass
3	API rate limits for the Pro plan?	1.00	1.00	1.00	1.00	Pass
4	How do I reset my password?	1.00	1.00	1.00	1.00	Pass
5	How do I cancel my subscription?	1.00	1.00	1.00	1.00	Pass
6	How do I enable 2FA?	1.00	0.00	1.00	0.67	Warn
7	How do I get a refund?	1.00	0.00	0.00	0.33	Drift

Rolling Average and Drift Analysis

The drift monitor computes a rolling 7-day average. The declining trend triggered by rows 6 and 7 demonstrates automated detection of retrieval degradation.

Faithfulness (avg)	**1.00**
Relevance (avg)	**0.71**
Resolution (avg)	**0.86**
Overall rolling average	**0.86**

Drift Analysis: The overall rolling average of 0.86 remains above the 0.70 alert threshold. However the declining trend (-0.33) was correctly identified — Relevance scores dropped sharply on the last two evaluations. In production, this would trigger a Slack/email alert prompting the team to review and re-ingest updated documentation.

Technical Implementation

Technology Stack

Component	Technology	Purpose
Document Loading	**LangChain DirectoryLoader**	Load .txt/.md help files
Text Splitting	**RecursiveCharacterTextSplitter**	Chunk docs (500 chars, 50 overlap)
Embeddings	**HuggingFace all-MiniLM-L6-v2**	Local free vector embeddings
Vector Store	**ChromaDB**	Persist and search embeddings
LLM (primary)	**groq/llama-3.1-8b-instant**	Answer generation (free tier)
Fallback	**Groq direct API**	If Respan gateway fails
Gateway	**Respan AI Gateway**	Model routing, tracing, caching
Tracing SDK	**respan-ai + instrumentation**	Auto-trace all LLM calls
Eval judges	**Groq LLM (3 prompts)**	Score faithfulness/relevance/resolution
Drift detection	**Custom Python (monitor.py)**	Rolling avg + threshold alerting
Language	**Python 3.12**	All pipeline components

Project File Structure

```
support-bot/ ■■■ src/ ■ ■■■ ingest.py # Load docs -> embed -> ChromaDB ■ ■■■
rag.py # Retrieve -> augment -> call LLM via Respan ■ ■■■ evaluate.py # 3x
LLM-as-judge scoring ■ ■■■ monitor.py # Rolling avg drift detection ■ ■■■ demo.py
# End-to-end demo runner ■■■ docs/ # Help documentation (.txt files) ■■■
.env.example # Environment variable template ■■■ requirements.txt # Python
dependencies
```

Run Commands

1. Ingest documents

```
python src/ingest.py
```

2. Ask a question

```
python src/rag.py "How do I reset my password?"
```

3. Evaluate response

```
python src/evaluate.py "How do I reset my password?"
```

4. Check for drift

```
python src/monitor.py
```

5. Full demo

```
python src/demo.py
```

Conclusion

This project demonstrates end-to-end AI engineering reflecting real production concerns — not just building a system that works, but one that stays reliable over time. Key differentiators:

Automated Quality Assurance	Every LLM response is evaluated by 3 independent judges without human intervention, making quality measurement systematic and scalable.
Proactive Drift Detection	The system monitors rolling quality trends and alerts the team the moment degradation begins, enabling pre-emptive remediation before users are affected.
Full Observability	Every LLM call is traced with Respan — latency, cost, tokens, and eval scores visible in one dashboard, enabling fast debugging and data-driven model selection.
Zero Infrastructure Cost	By combining Groq free tier with local HuggingFace embeddings, the entire pipeline runs at $0.00, demonstrating cost-conscious engineering.
Production Patterns	Model routing with fallback, prompt versioning, rolling metric windows, and CI-compatible exit codes reflect patterns used in production AI systems at scale.

GitHub	github.com/pranavshankar-dev/respan-support-bot
Respan Dashboard	platform.respan.ai | Project: AI Customer Bot Default
Respan Org ID	7d80e5ff-a37a-49ce-97d1-7e0cbc17c7a7